Program - 8

Demonstration of Data Engineering Processes

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Aim

Demonstrate data engineering processes by collecting data from multiple sources, integrating datasets, cleaning them, and storing the processed results.

Algorithm

1. Start
2. Import the pandas library.
3. Create the Student Data containing Student ID, Name, and Department.
4. Create the Marks Data containing Student ID and Marks.
5. Create the Attendance Data containing Student ID and Attendance.
6. Convert each data dictionary into separate Pandas DataFrames:
   - students
   - marks
   - attendance
7. Display the Student, Marks, and Attendance datasets.
8. Merge the Student Data with the Marks Data using Student_ID as the common key.
9. Merge the resulting DataFrame with the Attendance Data using Student_ID.
10. Display the integrated student dataset.
11. Check for missing values in the integrated dataset.
12. Remove duplicate records using drop_duplicates().
13. Replace missing values in the Marks column with 0.
14. Replace missing values in the Attendance column with 0.
15. Calculate the Performance Score using:
    - 70% of Marks
    - 30% of Attendance
16. Add the calculated value as a new Performance column.
17. Display the cleaned and transformed dataset.
18. Save the complete processed dataset as processed_students.csv.
19. Display a message confirming that the processed data has been saved.
20. Display the names of the output files.
21. Stop.

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```
import pandas as pd
  student_data = {
        'Student_ID': [101, 102, 103, 104, 105],
        'Name': ['Arun', 'Bala', 'Cathy', 'Divya', 'Ezhil'],
        'Department': ['CSE', 'ECE', 'CSE', 'EEE', 'ECE']
}
students = pd.DataFrame(student_data)
marks_data = {
        'Student_ID': [101, 102, 103, 104, 105],
        'Marks': [85, 72, 90, 65, 88]
}
marks = pd.DataFrame(marks_data)
attendance_data = {
        'Student_ID': [101, 102, 103, 104, 105],
        'Attendance': [95, 80, 92, 75, 90]
}
attendance = pd.DataFrame(attendance_data)
print("Student Data:")
print(students)
print("\nMarks Data:")
print(marks)
print("\nAttendance Data:")
print(attendance)
  combined = pd.merge(students, marks, on='Student_ID')
combined = pd.merge(combined, attendance, on='Student_ID')
print("\nIntegrated Data:")
print(combined)
print("\nMissing Values:")
print(combined.isnull().sum())
combined = combined.drop_duplicates()
combined['Marks'] = combined['Marks'].fillna(0)
combined['Attendance'] = combined['Attendance'].fillna(0)
combined['Performance'] = (
        combined['Marks'] * 0.7 +
        combined['Attendance'] * 0.3
)
print("\nAfter Cleaning and Transformation:")
print(combined)
combined.to_csv('processed_students.csv', index=False)
print("\nProcessed data saved successfully!")
print("Files created:")
print("1. processed_students.csv")
print("2. eligible_students.csv")
```